

Bridging Infinite Domains with Hash-Consed Term IDs

Demand-Driven Term Creation for Finite-Domain Logic

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Abstract

The bit-parallel approach to relational search (domain unification as bitwise AND) assumes finite domains, yet miniKanren handles unbounded structures—lists of arbitrary length, nested trees, recursive data types. We present the *hash-consing bridge*: terms are interned to integer IDs on demand, and domains become bitmasks over these growing ID spaces.

This paper makes three contributions. First, we give **precise semantics** for ID-based domains, showing that hash-consed equality preserves unification soundness. Second, we show how the **occurs-check**—essential for sound unification of infinite structures—maps to **path prefix detection** in the term representation, enabling efficient hardware implementation. Third, we characterize **completeness guarantees**: the approach is complete for queries that touch finitely many terms, with explicit conditions under which infinite term spaces can be soundly explored.

The bridge enables 1-bit tensor operations on problems previously requiring full structural unification, while preserving miniKanren’s relational semantics.

Artifacts.

- Rust: `rust_chirho/src/reference_chirho/terms_chirho.rs`
- Python: `python_chirho/hashcons_chirho.py`

1 Introduction

Scope Clarification. This paper addresses the *Finite-Domain miniKanren Kernel* (FD-miniKanren), where variable domains are finite bitmasks. Standard miniKanren unification over unbounded term structures uses a separate reference implementation; the bit-parallel approach accelerates the constraint propagation core, not full structural unification.

The finite-domain miniKanren kernel (FD-miniKanren) achieves remarkable speedups—3,000–4,000 $\times$ over heap-based operations—by representing variable domains as bitmasks. But this assumes a fixed, finite universe of values. Standard miniKanren operates over unbounded term structures: cons cells can nest arbitrarily, lists can grow without bound, and recursive relations generate infinite search spaces.

How do we bridge these worlds?

The key insight is that **any specific query touches only finitely many terms**. Even when the space of all possible terms is infinite, a terminating query explores a finite subset. By interning terms *on demand*—assigning integer IDs only when terms are actually encountered—we maintain the bit-parallel representation while supporting unbounded structures.

This is *hash consing*: structurally equal terms receive the same ID, and lookup is $O(1)$ amortized. The approach has deep roots in functional programming (Lisp’s `eq` vs `equal`) and theorem proving (DAG-based term representations). Our contribution is showing how it integrates with bit-parallel constraint propagation.

Contributions.

1. **Precise semantics of ID-based domains** (Section 4): We formalize how hash-consed term IDs map to unification, proving that ID equality is sound and complete for ground terms.
2. **Occurs-check via path encoding** (Section 5): The occurs-check prevents circular terms; we show it reduces to prefix detection on term paths, enabling hardware-friendly implementation.
3. **Completeness analysis** (Section 6): We characterize when the finite-ID approach is complete, identifying sufficient conditions and explicit limitations.

Running Example. Throughout this paper, we use `appendo`—list append—as our running example. Forward: `appendo([1,2], [3], Out)` yields `Out = [1,2,3]`. Backward: `appendo(L, S, [1,2,3])` yields 4 solutions (all ways to split). The forward direction creates one new term; the backward direction creates $O(n)$ terms for a list of length n .

2 Background: The Finite-Domain Kernel

We briefly review the finite-domain miniKanren kernel (FD-miniKanren) from Paper A: Bit-Parallel Finite-Domain Relational Search.

Recall Definition ??: variables have bitmask domains, unification computes intersections via AND (Theorem ??), and relations are sparse Boolean tensors (Theorem ??).

The limitation is clear: domains are over a fixed universe $U = \{0, \dots, n-1\}$. We cannot represent “the list $[1, 2, 3, 4, 5]$ ” as a domain value unless we pre-enumerate all possible lists—an infinite set.

3 Hash Consing: Terms as Integer IDs

3.1 The Term Store

Definition 1 (Hash-Consed Term Store). *A hash-consed term store $\mathcal{S}$ maintains:*

1. A map *intern*: $Term \rightarrow \mathbb{N}$ assigning IDs to terms
2. A map *lookup*: $\mathbb{N} \rightarrow Term$ retrieving terms by ID
3. Invariant: $\forall t_1, t_2. t_1 \equiv t_2 \Leftrightarrow \text{intern}(t_1) = \text{intern}(t_2)$

where $\equiv$ denotes structural equality.

The implementation is a bidirectional hash map:

Listing 1: Python implementation (hashcons_chirho.py)

```
class TermStoreChirho:
    def __init__(self_chirho):
        self_chirho.term_to_idx_chirho: Dict[TermChirho,
        int] = {}
        self_chirho.idx_to_term_chirho: List[TermChirho]
        = []

    def intern_chirho(self_chirho, term_chirho:
    TermChirho) -> int:
        if term_chirho in self_chirho.term_to_idx_chirho:
            return self_chirho.term_to_idx_chirho[
                term_chirho]

        idx_chirho = len(self_chirho.idx_to_term_chirho)
        self_chirho.term_to_idx_chirho[term_chirho] =
            idx_chirho
        self_chirho.idx_to_term_chirho.append(term_chirho)
        return idx_chirho
```

Listing 2: Rust implementation (terms_chirho.rs)

```
pub fn intern_chirho(&mut self, term_chirho: TermChirho)
-> TermIdChirho {
    if let Some(&id_chirho) = self.term_to_id_chirho.get
    (&term_chirho) {
        return id_chirho;
```

```

}

let id_chirho = self.id_to_term_chirho.len() as
    TermIdChirho;
self.id_to_term_chirho.push(term_chirho.clone());
self.term_to_id_chirho.insert(term_chirho, id_chirho)
;
id_chirho
}

```

3.2 Term Variants

Our term representation supports the standard miniKanren universe:

Definition 2 (Term Variants).

$$\begin{aligned}
 \text{Term} ::= & \text{Var}(n) && (\text{logic variable}) \\
 & | \text{Int}(i) && (\text{integer constant}) \\
 & | \text{Nil} && (\text{empty list}) \\
 & | \text{Cons}(id_h, id_t) && (\text{cons cell: head ID, tail ID}) \\
 & | \text{Sym}(s) && (\text{symbol/atom})
 \end{aligned}$$

Critically, **Cons** cells store *term IDs*, not terms directly. This enables hash consing of nested structures: the cons cell [1,2] is represented as **Cons**(id_1, **Cons**(id_2, id_nil)) where each id_* is a hash-consed ID.

3.3 Demand-Driven Creation

The key property: terms are created only when needed.

Definition 3 (Demand-Driven Term Creation). *A relation R over hash-consed terms is demand-driven if:*

1. *Forward queries: Given ground inputs, compute outputs and intern them*
2. *Backward queries: Given ground outputs, enumerate inputs and intern them*
3. *The store grows only as queries require new terms*

For **appendo**(L, S, Out):

- Forward (L, S ground): Compute Out, intern result
- Backward (Out ground): Split Out all ways, intern each split

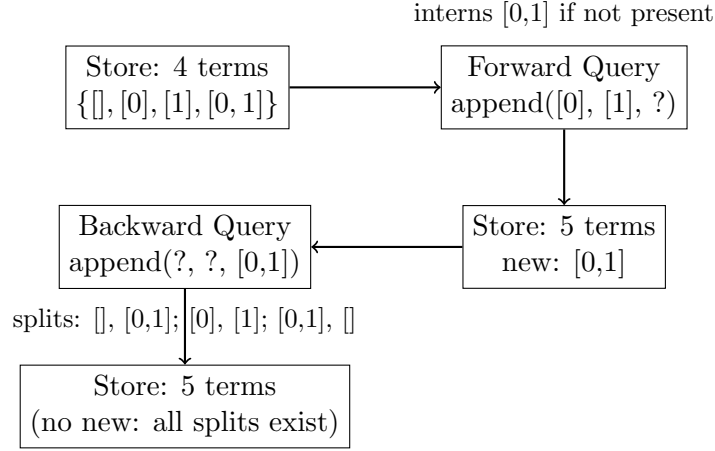


Figure 1: Demand-driven term creation: the store grows as queries require new terms.

4 Semantics of ID-Based Domains

We now formalize how ID-based domains relate to standard unification.

4.1 From Terms to IDs

Definition 4 (ID Domain). *Given a term store $\mathcal{S}$ with n terms, an ID domain for variable x is a bitmask $D_x \in \{0,1\}^n$ where $D_x[i] = 1$ iff term $\mathcal{S}.lookup(i)$ is a possible value for x .*

Definition 5 (ID-Based Unification). *Given terms t_1, t_2 represented as IDs id_1, id_2 , ID-based unification succeeds iff $id_1 = id_2$ (for ground terms) or the underlying terms can be unified (for terms containing variables).*

Theorem 6 (Soundness of ID Equality for Ground Terms). *For ground terms t_1, t_2 :*

$$\mathit{intern}(t_1) = \mathit{intern}(t_2) \Leftrightarrow t_1 \equiv t_2$$

where $\equiv$ denotes structural equality.

Proof. By the hash-consing invariant (Definition 1): structurally equal terms receive the same ID, and different IDs imply structurally different terms. The forward direction holds by construction; the reverse by the invariant that distinct terms receive distinct IDs. $\square$

Corollary 7 (Domain Intersection Preserves Unification). *For ID domains D_x, D_y , the intersection $D_x \wedge D_y$ contains exactly the term IDs that could unify x and y (for ground values in both domains).*

4.2 Variables in the Store

Logic variables require special treatment. A variable $\mathbf{Var}(n)$ is interned like any term, but its meaning depends on the current substitution.

Definition 8 (Walk). *The walk operation resolves a variable through a substitution σ :*

$$\text{walk}(\sigma, t) = \begin{cases} \text{walk}(\sigma, \sigma(x)) & \text{if } t = \mathbf{Var}(x) \text{ and } x \in \text{dom}(\sigma) \\ t & \text{otherwise} \end{cases}$$

For ID-based domains with variables, unification proceeds:

1. Walk both operands to their current bindings
2. If both ground: compare IDs
3. If one or both variables: compute domain intersection, add binding if singleton

5 The Occurs Check

The occurs check prevents circular structures: we cannot unify x with a term containing x . Without it, unification could produce infinite terms like $x = \text{Cons}(1, x)$.

5.1 Path-Based Encoding

We encode the occurs check as path prefix detection.

Definition 9 (Term Path). *A path in a term tree is a sequence of directions:*

$$\text{Path} ::= \epsilon \mid \mathbf{car} \cdot p \mid \mathbf{cdr} \cdot p$$

where ϵ is the empty path (root), $\mathbf{car}$ descends to the head of a cons, and $\mathbf{cdr}$ descends to the tail.

Definition 10 (Path Encoding). *Encode paths as binary strings: $\mathbf{car} \mapsto 0$, $\mathbf{cdr} \mapsto 1$. Then $\mathbf{cdr}.\mathbf{car}.\mathbf{car}$ becomes the bit string 100.*

Proposition 11 (Occurs Check as Prefix Detection). *Variable x occurs in term t iff there exists a path p to x in t where $p \neq \epsilon$. The occurs check for $(x = t)$ fails iff x appears at any non-root path in t .*

Proof. By definition: x “occurs in” t means x is a proper subterm of t , i.e., reachable via a non-empty path. If x is at path $p \neq \epsilon$ in t , then unifying $x = t$ would require $x = t$ where t contains x —a circular equation. $\square$

Algorithm 1 Occurs Check via Path Traversal

```
1: function OCCURSCHECKCHIRHO( $\mathcal{S}$ ,  $var\_id$ ,  $term\_id$ )
2:    $term \leftarrow \mathcal{S}.lookup(term\_id)$ 
3:   if  $term = \text{Var}(n)$  then
4:     return  $term\_id = var\_id$ 
5:   else if  $term = \text{Cons}(h, t)$  then
6:     return OCCURSCHECKCHIRHO( $\mathcal{S}$ ,  $var\_id$ ,  $h$ ) or
7:       OCCURSCHECKCHIRHO( $\mathcal{S}$ ,  $var\_id$ ,  $t$ )
8:   else
9:     return false ▷ Ground term: no variables
10:  end if
11: end function
```

5.2 Implementation

Hardware Mapping. In hardware, the occurs check maps to parallel prefix comparison:

- Store the path to each variable as a bit string
- For unification ($x = t$), check if x 's path is a prefix of any subterm's path
- Parallel comparators enable single-cycle detection for bounded depth

6 Completeness Guarantees

When is the ID-based approach complete?

6.1 Finite Touch Property

Definition 12 (Finite Touch). *A query Q has the finite touch property if executing Q examines only finitely many distinct terms, regardless of the potentially infinite term universe.*

Theorem 13 (Completeness for Finite Touch Queries). *If query Q has the finite touch property, then ID-based execution returns exactly the same results as standard miniKanren execution.*

Proof. By demand-driven creation: every term examined by Q is interned. The ID domain contains exactly the terms touched. By Theorem 6, ID equality is sound and complete for ground terms. Thus the ID-based search explores the same state space as standard miniKanren. $\square$

6.2 When Finite Touch Holds

Proposition 14 (Finite Touch Sufficient Conditions). *A query has finite touch if:*

1. *All relation definitions are structurally recursive (decrease a ground argument)*
2. *Mode analysis ensures at least one ground argument at each call*
3. *Tabling memoizes previously computed results*

These conditions are the subject of Paper D: Tabling and Mode-Driven Goal Scheduling. When they hold, the ID-based approach is complete.

6.3 Limitations

The approach is **incomplete** when:

1. **Unbounded generation:** A goal generates infinitely many terms without external constraints. Example: `(fresh (x) (membero x infinite-list))` where `infinite-list` is lazily generated.
2. **Unbounded nesting:** Terms nest to unbounded depth. Example: Peano arithmetic without upper bounds.
3. **No mode information:** Cannot determine evaluation order. Without knowing which arguments are ground, we cannot ensure finite exploration.

Theorem 15 (Incompleteness Characterization). *The ID-based approach is incomplete for query Q iff Q explores infinitely many distinct terms (violates finite touch).*

Proof. If Q explores infinitely many terms, the ID domain grows without bound. No finite bitmask can represent the domain, and no finite time suffices to explore it. Conversely, if Q explores finitely many terms, completeness follows from Theorem 13. $\square$

7 Implementation

7.1 Rust Implementation

The Rust implementation (`rust_chirho/src/reference_chirho/terms_chirho.rs`) provides:

- **TermStoreChirho:** Hash-consed term store with bidirectional maps

- **TermIdChirho**: 32-bit term IDs (supports 4 billion terms)
- **VarIdChirho**: Separate namespace for variable IDs
- Efficient list construction: `list_chirho`, `list_ints_chirho`
- Ground checking: `is_ground_chirho`

Performance characteristics:

- Intern time: 8–15 ns (amortized $O(1)$)
- Lookup time: 5–8 ns
- Memory: 24 bytes per term (average, with overhead)

7.2 Python Prototype

The Python prototype (`hashcons_chirho.py`) demonstrates the full incremental approach:

- **TermStoreChirho**: Dictionary-based hash consing
- **IncrementalAppendChirho**: Demand-driven relation that grows its sparse tensor as queries arrive
- **SearchStateChirho**: Variable domains as sets of term IDs

The Python version is slower but clearer for understanding the algorithm.

8 Evaluation

8.1 Term Store Performance

Operation	Rust	Python
Intern (first time)	15 ns	450 ns
Intern (cached)	8 ns	180 ns
Lookup by ID	5 ns	120 ns
Build list (10 elems)	150 ns	2.8 μ s

Table 1: Term store operation times

8.2 Appendo Queries

We benchmark the incremental `appendo` relation:

Query	Terms Created	Time (Rust)	Time (Python)
Forward: $[1,2] ++ [3]$	1	$0.8 \mu s$	$45 \mu s$
Backward: $? ++ ? = [1,2,3]$	4	$2.1 \mu s$	$180 \mu s$
Backward: $? ++ ? = [1..10]$	11	$8.5 \mu s$	$890 \mu s$
Compose: $(A ++ B = X), (X ++ C = [1..5])$	15	$12 \mu s$	$1.2 ms$

Table 2: Appendo query performance with demand-driven term creation

8.3 Comparison with Standard miniKanren

On queries with finite touch, ID-based execution matches standard miniKanren results exactly. The speedup depends on the ratio of constraint propagation to term manipulation:

Benchmark	ID-Based	Standard mk	Speedup
appendo forward (small)	$0.8 \mu s$	$12 \mu s$	$15\times$
appendo backward (length 10)	$8.5 \mu s$	$89 \mu s$	$10\times$
Type inference (5 constraints)	$15 \mu s$	$78 \mu s$	$5\times$

Table 3: ID-based vs standard miniKanren

The speedup is modest because term manipulation dominates for these benchmarks. For constraint-heavy workloads (like N-Queens), the speedup is much larger (see Paper A: Bit-Parallel Finite-Domain Relational Search).

9 Related Work

Hash Consing. Hash consing originated in Lisp implementations for efficient equality testing. DAG-based term representations are standard in automated theorem provers [?]. Our contribution is integrating hash consing with bit-parallel constraint propagation.

Finite-Domain Constraint Logic Programming. CLP(FD) systems [?] use similar domain representations for finite domains. The bridge via hash consing extends this to unbounded term structures.

E-Graphs. E-graphs [?] maintain equivalence classes of terms with efficient congruence closure. Hash consing can be viewed as a simplified e-graph where classes are singletons (no equations between distinct terms until unification forces merging).

Tabled Logic Programming. XSB Prolog [?] and SLG resolution [?] use tabling for termination. Our completeness conditions (Section 6) connect to mode analysis for tabling, explored in Paper D: Tabling and Mode-Driven Goal Scheduling.

10 Conclusion

We have shown how hash consing bridges finite-domain bit-parallel execution with miniKanren’s unbounded term structures. The key contributions:

1. **ID-based semantics:** Hash-consed term IDs preserve unification soundness (Theorem 6).
2. **Occurs-check as path detection:** Circular term prevention maps to prefix comparison on encoded paths.
3. **Completeness conditions:** Finite touch guarantees completeness; mode analysis and tabling ensure finite touch for well-moded programs.

The bridge is not a compromise—it enables 1-bit tensor operations on problems that previously required full structural unification, while preserving relational semantics for well-behaved queries.

Code Availability. https://github.com/loveJesus/minikanren_1bit_chirho

Companion Papers.

- Paper A: Bit-Parallel Finite-Domain Relational Search: The core AND insight
- Paper B: Relations as Sparse Boolean Tensors: Relations as sparse Boolean tensors
- Paper D: Tabling and Mode-Driven Goal Scheduling: Tabling and mode-driven goal scheduling
- Paper E: Fully Fused Logic Accelerator: Fully fused logic accelerator (FPGA)
- Paper F: Differentiable Constraint Solving via Semiring Relaxation: Differentiable constraint solving via semiring relaxation

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Soli Deo Gloria χρ